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Leveraging Digital Technology to Improve Monitoring and Planning in Public Sector Supply Chains: Evidence From India's Food Security Program

Maya Ganesh[1], Sarang Deo[2] and Sripad K Devalkar[2]

Abstract

Governments across the developing world are adopting digital technologies to improve the performance of public sector supply chains. However, empirical evidence regarding the impact of these technologies on performance is mixed. In this paper, we evaluate the impact of using biometric authentication (BA) devices on the performance of India's food security program. The program aims to deliver subsidized food grains to nearly 160 million low-income households through approximately 500,000 fair price shops (FPSs). Nearly half of those grains do not reach the beneficiaries and are diverted to the open market for private benefit. We estimate the impact of installing BA devices on reducing diversion in more than 3,300 FPSs in the Indian state of Karnataka between 2013 and 2015 using a difference-in-differences approach. We find that BA devices reduce up to 4% of the current diversion at the last mile. We argue that this relatively low impact is due to the presence of a workaround mechanism which results in imperfect monitoring and validate the decrease in diversion mechanism by estimating the differential impact of BA devices. We complement the empirical analysis with a simulation exercise to estimate the additional value that can be unlocked by using the BA devices. We find that the value of better replenishment planning using sales and inventory information captured by BA devices can be significant. The value of better planning can be as high as six times the value of reduction in diversion, depending on the preintervention levels of demand from genuine beneficiaries and diversion at the FPSs.

Keywords

Digital technology, food security program, public sector supply chains, value of information, value of monitoring

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1 Introduction

Public sector supply chains in many developing countries routinely experience diversion of subsidized goods to private markets across a wide range of sectors such as food and nutrition, health, fuel, and agricultural inputs (The Hindu, 2015; Khera, 2011; Saha, 2015). For instance, only 50% of the food grains that enter the Indian food security program's supply chain (known as the public distribution system [PDS]) comprising 160 million households and 500,000 fair price shops (FPSs) reach the targeted beneficiaries (Government of India, 2015). Incentive and/or audit contracts used for monitoring quality and regulatory compliance in private sector supply chains (e.g., Plambeck and Taylor, 2015; Cho et al., 2019) may not be feasible or effective in these settings due to: (i) the scale and the distributed structure of their operations, (ii) the political influence wielded by the supply chain players to evade monitoring, and (iii) the slow and ineffective legal enforcement of penalties arising from violations discovered during audits.

Recent advances in digital technology offer alternative, cost-effective monitoring approaches to reduce the extent of

[1]Operations and Decision Sciences, Indian Institute of Management Ahmedabad, Ahmedabad, Gujarat, India
[2]Operations Management, Indian School of Business, Hyderabad, Telangana, India

Corresponding author:
Maya Ganesh, Operations and Decision Sciences, Indian Institute of Management Ahmedabad, Vastrapur, Ahmedabad, Gujarat 380015, India.
Email: mayag@iima.ac.in

diversion in such contexts. For instance, biometric authentication (BA) devices that require validation of the identity of beneficiaries have been installed at thousands of FPSs across India to prevent fraudulent sales transactions by FPS dealers (Masiero, 2017; Allu et al., 2019). However, the potential value of better monitoring through these technologies may not be fully realized if associated processes are not appropriately designed. For instance, allowing a manual workaround (paper-based method) in the event of failure of BA devices (e.g., heat, dust, interruptions in power, and network availability) may be exploited by FPS dealers to continue diversion (Sachdev, 2018).

In this paper, we empirically quantify the value of using BA devices to curb diversion in the PDS, in the state of Karnataka. BA devices were installed in more than 3,300 FPSs in a phased manner between January 2013 and December 2015. We use publicly available data and conduct a quasi-experimental (difference-in-differences) analysis to compare the change in recorded sales in FPSs receiving a BA device with similar FPSs that did not receive one. Since diversion (or its change) cannot be directly observed, we use the change in recorded sales as its proxy. Based on our field observations and interviews, we believe that this is appropriate as the workaround mechanism (paper-based method) described above was reasonably effective in ensuring that the installation of BA devices did not affect genuine sales to beneficiaries. We substantiate this reasoning by providing empirical results on mechanisms that are likely to affect diversion but not genuine sales in Section 6. We find that BA devices reduced diversion on average by 36.3 kg per FPS per month. This translates to a reduction of up to 4% of the current diversion at the last mile. We also find that the impact of BA devices is lower (i) for FPSs located in regions with a higher proportion of economically and socially vulnerable population groups, (ii) for FPSs that are farther away from markets where dealers can dispose of diverted grains, and (iii) in periods with higher (than median) open market prices.

The reduction in diversion translates to ≈36.3 million INR per year (≈USD 550,000) in terms of monetary value of better monitoring (when calculated for the 3,075 FPSs which received BA devices in our analysis data set). Extrapolated to the entire state (23,241 FPSs), this would result in savings of ≈301.2 million INR per year (≈4.6 million USD). Combined with the investment required for installation, these savings yield a payback period of under two years.

In our study setting, the BA devices were equipped with the additional capability to record real-time information on sales and inventory at the FPSs. This timely information provided by the BA devices, which was not available in the erstwhile manual process, was, however, not used by the central planners to make the monthly inventory replenishment decisions. We undertake an extensive simulation study to estimate the potential incremental value from using the timely sales and inventory information for planning, which we define as value of better planning. This study is calibrated using both the structure of our empirical data and the results from our empirical study, supplemented with other operational data from multiple Indian states. Our estimation reveals that the average incremental value of better planning ranges between 309% and 557% of the value of monitoring and is particularly high when the average demand from genuine beneficiaries and the average diversion level before BA installation is neither too high nor too low. Notably, a significant portion of this value stems from the reduction in open market purchases by households due to reduced FPS stock-outs, a benefit that cannot be achieved through better monitoring alone.

Findings from our econometric and simulation analysis have important implications for the implementation and evaluation of digital technologies in large public welfare programs. First, although the main motivation for implementing these technologies is to improve monitoring, policymakers should be cognizant that information captured by these devices can be used to improve operational planning. This is true in many public welfare programs because both inadequate monitoring and inefficient planning can be attributed to the same root cause—lack of timely and accurate operational information—which is resolved by the new technology. Second, conventional impact evaluation approaches (Gertler et al., 2016) should be complemented with careful operational analysis to understand the relative magnitude of these two drivers of value. Apart from the immediate policy relevance, our findings advance the discussion at the confluence of two streams of supply chain literature: (i) emerging analytical work that proposes incentive contracts to improve socially responsible behavior in decentralized supply chains (e.g., Cho et al., 2019; Babich and Tang, 2012) and (ii) well-established analytical work that characterize the benefits of sharing information in decentralized supply chains (e.g., Lee et al., 1997, 2000).

We expand on these contributions to the relevant streams of prior literature in Section 2. We describe the empirical setting of PDS in Karnataka and the specific implementation of BA devices in Section 3. The accompanying data and our econometric approach to estimate the value of better monitoring using BA devices are described in Section 4. We present our main results in Section 5. In Section 6, we estimate heterogeneity in the impact of BA devices on different categories of FPSs to test the different mechanisms that affect diversion. In Section 7, we describe the simulation study to estimate the value of better planning using timely inventory information provided by BA devices. Finally, Section 9 provides concluding remarks.

2 Literature Review

Our work contributes to three substreams of the supply chain management literature.

2.1 Effectiveness of Incentive and Monitoring Contracts

The first, emerging, stream of work develops analytical models to evaluate the (in)effectiveness of various incentive and monitoring contracts in making supply chain partners comply with ethical and environmentally sustainable practices. Cho et al. (2019) evaluate how pricing and inspection strategies can be used to restrict a supplier's use of child labor and find that reducing the cost of inspection does not necessarily reduce the use of child labor. Plambeck and Taylor (2015) show that increased audit frequency of supplier facilities can backfire, that is, may reduce supplier's compliance effort and increase effort toward hiding noncompliance to prevent detection during an audit. Huang et al. (2022) consider managing social responsibility in a three-tier supply chain where Tier 2 suppliers can potentially violate standards. They characterize conditions for optimality of either delegation (only Tier 1 supplier works with Tier 2 supplier) or control (the buyer or Tier 0 works directly with Tier 2) strategies. They also show that external pressure from nongovernmental organizations or the government on Tier 0 (buyer) or Tier 1 suppliers can sometimes backfire and lead to lower levels of social responsibility.

A related substream analyzes incentive contracts to deter product adulteration by suppliers. Mu et al. (2014) investigate quality incentives in the context of milk supply chains in emerging markets and show that confessor rewards for low-quality producers can improve milk quality and station profits, while quality rewards for high-quality producers may be counter-productive. Another study by Mu et al. (2016) within the same context shows that subsidizing the cost of product testing or investing in better testing infrastructure can lead to lower quality due to competition among milk suppliers and consequent free-riding. Levi et al. (2020) also study the impact of the accuracy of testing on adulteration decisions in the presence of exogenous quality uncertainty in a distributed food supply chain. Babich and Tang (2012) show that a deferred payment contract, wherein a part of the payment to the supplier is made with delay only if no adulteration is discovered by the customers, can perform better than the conventional inspection mechanism in deterring product adulteration. Rui and Lai (2015) extend this analysis when the procurement quantity is endogenously determined along with the optimal contract.

Our work complements this literature along three dimensions. First, in contrast to the existing literature that focuses on monitoring and auditing of suppliers, we focus on retailers. This increases the scale of the problem substantially (many more retailers than suppliers) making audits and inspections prohibitively expensive. Second, misaligned incentives in our context lead to quantity distortion rather than quality distortion. As a result, inspection mechanisms proposed in the literature to address quality distortion may not be applicable in our context. Third, barring a few exceptions (e.g., Mayer et al., 2004), the existing literature is focused on developing qualitative insights based on analytical models whereas our focus is on rigorous empirical quantification. Specifically, we measure the value of installing a BA device at the last mile delivery (FPS) and show a statistically significant impact in reducing diversion. We also find that workarounds can reduce the ability of BA mechanisms to reduce leakages. Our findings show that the value of an imperfect monitoring (owing to the coexisting exception handling mechanism) on supply chain performance in terms of leakage is positive, and indicates that there is value, even if monitoring is imperfect.

2.2 Bullwhip Effect and Value of Information

The second, well established, stream of literature is comprised of analytical, empirical, and simulation models that quantify the value of using downstream information for better upstream decision-making in supply chains to overcome the "Bullwhip effect" (Lee et al., 1997; Chatfield et al., 2004). Lee et al. (2000) find that a manufacturer can reduce its inventory cost by using downstream sales and inventory information shared by the retailer in addition to the retailer's order quantity, especially when demand is correlated over time. Subsequent papers show that a large part of this incremental value of shared information can be extracted from historical data on retailer's orders but the magnitude of this effect depends on the autoregressive structure of the demand distribution (Raghunathan, 2001; Gaur et al., 2005). Cachon and Fisher (2000) investigate various sources of value from implementing information technology in supply chains. They conducted a numerical study to show that the value from improved decision-making based on timely sharing of inventory and demand information is much lower than that from a reduction in lead time and batch sizes. In contrast, Cui et al. (2015) use data from a consumer packaged goods company to empirically show that a substantial reduction in forecast errors of future orders can be achieved if historic order data is augmented with downstream sales data from retailers. The main driver behind this improvement is that the retailer does not follow the optimal inventory policy, as is typically assumed in the literature, but deviates systematically from it and these deviations are further propagated in the ordering data.

Our empirical context differs significantly from previous studies in a few crucial aspects. The upstream echelon in our supply chain already has visibility into actual sales at the downstream echelon, albeit delayed, and makes all replenishment decisions based on them instead of retailer orders. Thus, the value of timely information (for increasing retail sales and reducing inventory) estimated in our analysis is over and above that captured in the existing literature on decentralized supply chains. Our study shows that even in a centralized supply chain, such as the PDS, using timely information on sales (or, ending inventories) for planning replenishments can unlock substantial value.

2.3 Technology Applications and Operational Benefits

Third, there is extensive literature on how technology applications have led to multiple operational benefits. For instance, radio frequency identification (RFID) and detection technology is used to ensure inventory accuracy. Inventory inaccuracy occurs when the inventory shown in the information system is not in agreement with the available inventory. Another related set of studies explores replenishment policies that account for inventory inaccuracy and investigate the value of RFID technology on lowering inventory shrinkage and achieving precise inventory replenishment due to accurate inventory visibility (Rekik, 2010; Fan et al., 2015; Lee et al., 2004; De Kok et al., 2008). There is a vast literature that studies returns from information technology investments in terms of decreased effort in information recording and processing, ease of computing, and ease of communication (Dewan and Kraemer, 2000; Kelley, 1994; Brynjolfsson and Hitt, 1996; Brynjolfsson and Yang, 1996; Stiroh, 2001; Rawley and Simcoe, 2013). Another set of related studies has explored the value of IT technologies such as RFID and micro electro mechanism system (MEMS) in improving tracking (Smith, 2014; Saygin, 2007; Doerr et al., 2006).

The technology adopted in our context has monitoring capabilities and hence implications for operational metrics such as diversion. The BA devices implemented in our context also have the additional capability of recording and storing real-time sales information. This dual role of the BA devices adds value on two different dimensions—value from monitoring by reducing leakages and value from planning due to the availability of timely information.

We estimate the value of technology-based interventions in the context of public sector operations in a developing economy. The findings from previous literature (Smith, 2014; Saygin, 2007; Doerr et al., 2006; Rekik, 2010; Fan et al., 2015; Lee et al., 2004; De Kok et al., 2008), which has primarily considered the use of technology in private sector supply chains in developed countries, do not directly translate to our context because developing economies often have weak technology infrastructure. Thus, technology-based interventions such as BA devices are often accompanied by workaround mechanisms that may be used, for example, the paper-based transaction recording that existed in parallel in our context, which reduces their effectiveness.

3 Empirical Setting: PDS Operations in Karnataka

We study PDS operations in the state of Karnataka during the period September 2013 to December 2015. During this period, the PDS in Karnataka served close to 14 million households through a network of more than 23,000 FPSs, across 35 districts. Roughly half of all the FPSs were operated by licensed private individuals while the rest were operated by cooperative societies (GOK, 2015).

Each FPS on average served 600 households. Each PDS beneficiary household was classified into one of two beneficiary categories—Antyodaya Anna Yojana (AAY) and Priority Households (PHH). Households that faced extreme economic vulnerability were classified as AAY households and were entitled to 35 kgs per month per household. Households under the PHH category consisted of all other low-income households unless they met specific exclusion criteria based on their socioeconomic characteristics (e.g., ownership of assets and land) and were entitled to 5 kg of subsidized grains per month per household member (Department of Food and Public Distribution, India, 2013). Each household was issued a ration card that established its identity and beneficiary category, and assigned to an FPS from where it could collect its monthly entitlement. Rice, the staple cereal in Karnataka, contributed 75%–85% of the total volume of entitlements, while wheat constituted the remainder. Eligible households were entitled to their quota of these two cereals at a rate of INR 3 and INR 2 per kg, respectively (Raju et al., 2018; NCAER, 2015). In addition to the regular entitlements, occasionally other commodities such as millets, pulses, sugar, and oil were also distributed through the PDS.

Next, we describe the process of sales, replenishment, and diversion at the FPSs (Sections 3.1 and 3.2), the process of auditing and monitoring FPSs' performance (Section 3.3) and the impact of the intervention, that is, installation of BA devices, on these processes (Section 3.4). The details in the following subsections are based on information gathered from publicly available sources as well as interactions with various stakeholders during field visits to different locations in Karnataka (details of the field visits are given in Appendix A in the E-Companion).

3.1 Sales and Replenishment Processes at the FPSs

At the beginning of each month, indexed by t, the Department of Food, Civil Supplies and Consumer Affairs (DFCSCA) provides each FPS, indexed by i, the list of beneficiary households that are eligible to collect their entitlements from that FPS, that is, the eligibility list. During the course of the month, beneficiaries come to their assigned FPS to collect food grains, primarily rice, at subsidized prices. The FPS dealer verified the details of the beneficiaries who came to the shop by matching the details on the ration card issued to the beneficiary against the eligibility list and issued grains. The sales transactions were recorded manually in a register, after acknowledgment from the beneficiary. The transaction was also recorded in the beneficiary's ration card.

The total quantity of grains disbursed from the FPS during the month constituted the recorded sales, $S_{i,t}$, for FPS i for month t. At the end of the month, a copy of all sales transactions recorded in the register along with the beneficiaries' acknowledgments on the eligibility list was collected from all FPSs. The paper records across all FPSs were consolidated, digitized, and uploaded to a central database in the DFCSCA headquarters by the middle of month $t + 1$.

All FPSs were replenished monthly, and the replenishment quantity was determined centrally. The replenishment quantity for FPS i for month $t + 1$, Q_{t+1}, was determined at the end of month t such that the inventory at the beginning of month $t+1$ equaled gross requirement, $\mathrm{GR}_{i,t+1}$, that is, the sum of entitlements of beneficiaries eligible to collect their grains from FPS i in month $t+1$. However, delayed processing of physical sales transaction records meant information about inventory at the end of month t, $\mathrm{CB}_{i,t}$ (closing balance), was not available at the time of determining the replenishment quantity. As a result, the ending inventory for month $t - 1$ was used as a proxy to calculate the replenishment quantity, that is, the replenishment quantity $Q_{i,t+1}$ was set equal to $\mathrm{GR}_{i,t+1} - \mathrm{CB}_{i,t-1}$. The replenishment quantity for all FPSs is calculated by the central team and FPSs do not decide the order quantity. Our understanding of the replenishment policy is based on our multiple field visits and discussions with the central planning team at the Bangalore office. For details on field visits, please refer to the Appendix in the E-Companion.

3.2 Diversion of Grains at the FPS

The FPS dealers were paid a commission of approximately INR 1 per kg of grains distributed to eligible beneficiaries. The commission rate was substantially lower than the market price of food grains (≈INR 25 to 30 per kg). This created a strong incentive for the FPS dealers to exploit the weakness in the manual process of recording sales and divert food grains away from eligible beneficiaries to the open market for private gains. The strong incentive combined with asymmetric power dynamics between the beneficiaries and the FPS dealers (who wield considerable local political influence) created an opportunity for the FPS dealers to divert food grains by exploiting loopholes in the manual record-keeping process. That is,

$$S_{i,t} = \tilde{S}_{i,t} + L_{i,t}, \tag{1}$$

where $\tilde{S}_{it}$ and L_{it} represent the quantity disbursed to genuine beneficiaries and quantity diverted, respectively, by FPS i in month t, neither of which is observed directly.

Our interviews with beneficiaries during the field visits revealed that FPS dealers use a variety of mechanisms to divert grains for private benefit. These include: (i) distributing less than the recorded quantity to beneficiaries who transacted in a given month, (ii) recording false transactions against beneficiaries who did not transact in a given month, and (iii) recording transactions against fraudulent ration cards that do not belong to any genuine beneficiary (NCAER, 2015; Dreze and Khera, 2015). In other words, the recorded sales $S_{i,t}$ is the sum of quantity disbursed to genuine beneficiaries, $\tilde{S}_{i,t}$, and the quantity diverted $L_{i,t}$ by FPS i in month t.

3.3 Monitoring of FPSs

FPSs were audited periodically by food inspectors to monitor and curtail the diversion of grains. The audit involved verification of sales transaction entries in the ration cards of beneficiaries and acknowledgment of the same in the FPS's records, and physical verification of food grain inventory in the FPS. Penalties for fraudulent activities (such as black marketing, overcharging, diversion, and under-scaling) by an FPS dealer can range from a monetary fine to the suspension of the FPS license (GoK 2016, GoAP 2018, GoJK 2023). Our interviews revealed that each food inspector was responsible for auditing 50 to 60 FPSs throughout the year. Given the labor-intensive nature of these audits, each food inspector was able to audit only four to five FPSs on average each month, and as a result, each FPS was audited only once about each year. The manual nature and low frequency of audits significantly limit their ability to detect diversion at the FPSs.

3.4 The Intervention: Installation of BA Devices at the FPSs

Starting in January 2013, BA devices were installed in a phased manner in 3,332 of the 7,313 FPSs in 11 districts (Figure A1 in the Appendix in the E-Companion). The proportion of FPSs receiving the BA devices varied significantly across districts (7% to 84%) and the devices were operational till December 2015. DFCSCA officials indicated that FPSs receiving the BA devices were chosen based on the availability of power, transportation, and telecommunication (mobile and internet) connectivity, and proximity to taluk (subdistrict) headquarters.

BA devices were installed with the aim of reducing FPS dealers' ability to divert food grains by recording fraudulent sales transactions. The BA device at each FPS had an electronic database of beneficiaries eligible to collect their entitlements at the FPS, their biometric identifiers (e.g., fingerprints), and entitlement details. The device was equipped to capture the physical biometrics of the beneficiary at the time of the transaction. The physical biometrics were compared against those stored in the device and authenticated. A sales transaction was recorded only upon successful verification.

To ensure that the installation of BA devices did not adversely affect genuine sales, FPS dealers were allowed to continue using the traditional paper-based sales register and eligibility list as a workaround mechanism to process and record transactions when the BA devices faced technical issues. These technical issues included poor quality of biometric images stored in the device, sensor malfunction due to exposure to heat and dust or power failures, and worn out or temporarily compromised physical biometrics due to age and/or heavy manual labor (Rosamma, 2016; Lakhani, 2018). During our interviews with beneficiaries during the field visits, many of them indicated the existence of such issues and FPS dealers using the workaround mechanism (paper-based method) to record transactions.

Furthermore, the roles and responsibilities of food inspectors and the audit process remained largely unchanged after the installation of BA devices: (i) verifying sales transactions in the ration cards of randomly chosen beneficiaries and cross-checking the recorded sales, captured through both the BA devices as well as paper-based transactions and (ii) physical verification of food grain inventory at a randomly chosen sample of FPSs. There was no change in the frequency of audits following the implementation of BA devices, nor an increased penalty for fraudulent transactions after the intervention. The manual and labor-intensive nature of the audit process continued to pose challenges in effectively detecting diversion at the FPS.

In addition to BA, the BA devices also stored and transmitted the sales transactions electronically to a central database thereby enabling real-time monitoring of inventory at the FPS. However, to account for transactions that used the workaround mechanism and were recorded manually, and those at FPSs without BA devices, the upstream planning and replenishment policy was not modified to utilize the updated information. Replenishment quantities continued to be determined using lagged inventory information as described in Section 3.1.

4 Data and Econometric Approach

4.1 Data Description

We use publicly available data related to PDS operations in 11 districts of Karnataka during the period from September 2013 to December 2015. Each record in our data is an FPS–month with the following information: (i) gross requirement at the FPS for that month for the AAY and PHH household categories, (ii) ending inventory of rice and wheat for the AAY and PHH households, and (iii) the sales transaction transmission status indicating whether a BA device was used to record sales transactions at a particular FPS in that particular month. For a given FPS, we consider the first month with an active transmission status as the month of installation of a BA device at that FPS.

After excluding observations with missing data and outliers, we obtained an unbalanced panel data set comprising 153,028 FPS–month observations corresponding to 6,934 FPSs (of which 3,075 FPSs had BA devices) over a period of 28 months. Table 1 provides summary statistics on key operational variables of interest.[1] We find that PHH beneficiaries contributed close to 90% of the gross requirement, while rice constituted 80% of the volume of PDS beneficiary entitlements across all household categories. Close to 60% of the FPSs in our data were in rural locations, while 45% of all the FPSs were managed by cooperative societies.

We calculated the recorded sales from the available data for each FPS for each month as follows:

$$S_{i,t} = CB_{i,t-1} + Q_{i,t} - CB_{i,t} = CB_{i,t-1} + GR_{i,t} - CB_{i,t-2} - CB_{i,t}, \quad (2)$$

where $CB_{i,t}$ is the inventory at FPS i at the end of month t, and $Q_{i,t}$ is the replenishment quantity received at the beginning of month t by FPS i. As described in Section 3.1, the replenishment quantity is equal to $\mathrm{GR}_{i,t}$ net of the lagged ending inventory $\mathrm{CB}_{i,t-2}$.

Overall, recorded sales constituted 99% of gross entitlements on average, with <1% remaining as inventory at the end of the month.

4.2 Econometric Approach

We use difference-in-differences (DID) methodology (e.g., Imbens and Wooldridge, 2009; Angrist and Pischke, 2008; Lechner et al., 2011) to compare the change in recorded sales at the FPS–month level between the treatment (FPSs with BA devices) and control (FPSs without BA devices) groups to estimate the impact of installing BA devices on diversion at the FPSs.

As described by equation (1), the recorded sales S_{it} is a sum of actual sales to genuine beneficiaries $\tilde{S}_{it}$ and the quantity diverted L_{it}, and the change in recorded sales due to installation of BA devices is the sum of the change in genuine sales and diversion, that is,

$$\Delta S_{it} = \Delta \tilde{S}_{it} + \Delta L_{it}.$$

From the data, we only observe S_{it} but not the components, $\tilde{S}_{it}$ and L_{it} and as a result cannot observe ΔL_{it}, the quantity of interest. However, we believe the change in recorded sales is primarily due to a change in diversion and not change in genuine sales for the following reasons: (i) the workaround mechanism (paper-based method) ensured that beneficiaries continued to receive the grains they would have received before BA devices were implemented in the event of problems with the BA devices and BA device failures are unlikely to result in a reduction of genuine sales to beneficiaries and (ii) the replenishment policy used through the period of our study was not affected by the implementation of BA devices, ensuring changes in recorded sales is not due to change in availability of food grains at the FPSs. In other words, the mechanism through which BA devices affect recorded sales is by affecting diversion, and not genuine sales to beneficiaries.

Our identifying assumption is that

> genuine sales to beneficiaries are not affected by the installation of BA devices and recorded sales at FPSs that are not installed with BA devices in a given month are a valid counterfactual for the recorded sales that would be obtained in the FPSs with BA devices in the absence of the intervention.

We present additional analysis in Section 6 in support of our claim that the primary mechanism through which BA devices affect recorded sales is through their impact on diversion, and not on genuine sales.

Table 1. Summary statistics on key operational variables of interest.

	Household category		FPS location		FPS ownership		Grain type		
	AAY	PHH	Urban	Rural	Private	Co-op	Rice	Wheat	Total
Gross requirement (MT)	1.36	10.50	9.83	13.01	11.01	12.74	9.33	2.54	11.86
	(1.29)	(7.50)	(8.37)	(7.87)	(7.11)	(9.33)	(6.48)	(2.45)	(8.23)
Recorded sales (MT)	1.35	10.44	9.66	13.01	10.90	12.72	9.27	2.52	11.79
	(1.29)	(7.48)	(8.3)	(7.84)	(7.08)	(9.29)	(6.45)	(2.45)	(8.19)
Ending inventory (kg)	5.28	49.81	77.20	40.60	46.25	65.35	40.43	14.66	55.09
	(49.01)	(241.93)	(298.11)	(231.10)	(190.67)	(325.81)	(186.43)	(140.76)	(259.69)
Observations	153,028	153,028	59,085	93,943	85,079	67,949	153,028	153,028	153,028

Notes: FPS = fair price shop; AAY = Antyodaya Anna Yojana; PHH = priority households; (1) MT denotes metric tonne (=1,000 kg). (2) All numbers for gross requirement, recorded sales, and ending inventory represent averages over all observations at the FPS–month level. (3) Numbers in brackets denote standard deviation values.

As mentioned in Section 3.4, the selection of FPSs into the treatment group was based on the availability of electricity, transportation and telecommunication connectivity, and proximity to Taluk headquarters. Hence, the composition of treatment and control groups may be different, and a direct comparison of $S_{i,t}$ between these groups may not capture the true treatment effect. We use a two-step approach that accounts for the difference in pretreatment characteristics of FPSs across the two groups to obtain an unbiased estimate of the treatment effect (Austin, 2013; Rosenbaum and Rubin, 1983). In the first step, we estimate the probability that an FPS will be selected for installation of a BA device (a propensity score), and assign weights to each observation in our data based on the FPS's propensity score. In the second step, we estimate the DID model using the weighted observations across the two groups. This two-step procedure yields a "doubly robust estimator" that remains unbiased even if one of the models is misspecified (Słoczyński and Wooldridge, 2018).

We find that nearly 99% of all observations fall in the range of propensity scores that is common to both control and treatment groups (0.014 to 0.99) and there is a substantial reduction in the values of the standardized differences between the control and treatment groups after weighting of observations. The reduction in standard differences indicates that the weighting procedure improves the comparability of the treatment and control groups on observed covariates and the ability to draw causal inferences about the treatment effect. The details of the propensity score estimation and weighting procedure are given in Appendix C in the E-Companion.

5 Value of Better Monitoring Using BA Devices

We normalize the recorded sales by the gross requirement for the month for each FPS i for each month t to make FPSs of different sizes comparable and use $\mathrm{SGR}_{i,t} = S_{i,t}/\mathrm{GR}_{i,t}$ as the dependent variable to estimate the average treatment effect using the following DID specification:

$$SGR_{i,t} = \alpha_i + \beta_t + \delta \times BA_{i,t} + \varepsilon_{i,t}, \tag{3}$$

Table 2. Overall impact of installing BA devices on curbing diversion of rice.

	Overall impact
Overall impact (δ)	-0.0039^{**}
	(0.0017)
R^2	0.167
Adjusted R^2	0.128
Observations	152334

Notes: BA = biometric authentication; FPS = fair price shop. (1) Results shown are for FPS–month panel regressions with FPS and month fixed effects and standard errors clustered at the month and taluk levels. (2) $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$. (3) Column (1) shows the results from estimating equation (3) with the dependent variable $\mathrm{SGR}_{i,t}$.

where $\mathrm{BA}_{i,t} = 1$ if a BA device was in use at FPS i in month t and 0 otherwise.

In equation (3), α_i and β_t capture time-invariant FPS and FPS-invariant time-fixed effects, respectively. Time-invariant FPS fixed effects include effects due to the location and accessibility of the FPS, the attitude of the FPS dealer, etc., while shop-invariant month fixed effects include effects such as festival seasons, agricultural activity, and seasonal migration for work. The average treatment effect, change in diversion as a fraction of the gross requirement due to BA installation, is given by δ. We calculate cluster-robust standard errors (Bertrand et al., 2004) with two-way clustering at the taluk and month level to account for correlation in the unobserved household (e.g., dietary preferences), administrative (e.g., inspection intensity), and market (e.g., prices) characteristics that vary over time and geography.

Estimating the above model with recorded sales and gross requirements for rice, we find that $\delta = -0.0039$, and is significant (Column (1) in Table 2). That is, recorded sales, and hence diversion was lower by $0.0039 \times \mathrm{GR}$ on average per month at an FPS post installation of a BA device.[2] To put the impact of BA devices on diversion in context, the extent of diversion at the FPS level in Karnataka was estimated to be 9.5% of the total entitlements (NCAER, 2015), implying BA devices were able to reduce the baseline diversion quantity by about 4% (= 0.0039/0.095).

A possible explanation of the relatively low impact lies in the capabilities of the BA devices and the mode of their implementation. The asymmetric power dynamics between FPS dealers and beneficiaries allowed the FPS owners to exploit the manual mechanism (which was continued to ensure genuine beneficiaries were not denied their entitlements due to technical issues) to record false transactions and continue diverting grains. In fact, many beneficiaries reported during our interviews that FPS dealers routinely used the workaround mechanism (paper-based method) citing device breakdowns, maintenance, and power issues even after the introduction of BA devices (see Appendix A in the E-Companion).

Despite their limited relative impact, technology-based interventions such as BA devices can still add substantial value given the scale of the PDS. Consider the case of the PDS in Karnataka. With an average monthly gross requirement of 9.32 MT (i.e., 9,320 kg) of rice per FPS per month and 3,075 FPSs with BA devices installed, a reduction of diversion by 0.39% of gross requirement translates to a reduction of 111.77 MT per month in absolute quantity. The reduction in diversion represents a reduction in total outflow of grains from the PDS, and therefore a reduction in the procurement requirements to the same extent. The incremental value of the reduction in procurement cost is approximately INR 36.3 million per year (≈USD 0.56 million per year at an exchange rate of ≈65 INR per USD).[34] The potential value of monitoring, if the intervention had been implemented in the entire state of Karnataka (23,241 FPSs and average gross requirement of 10.25 MT of rice), would have been USD 4.6 million. Accounting for the cost of implementation of around INR 20,000 per device (Business Today, 2016), these savings translate to a payback period of <21 months. In addition to the direct savings due to lower procurement, reducing diversion, however, small, also improves service to beneficiaries resulting in added value.

We conduct falsification and robustness checks to rule out the possibility that observed effects are due to chance. We also perform additional analyses to test the robustness of our results to changes in variable definitions, matching techniques, and to rule out other plausible explanations such as anticipatory behavior of FPS dealers (divert more in the immediate few months before implementation of BA device). The details of these tests and their results are described in Section F of the Appendix in the E-Companion.

In the next section, we present additional empirical evidence in the form of different heterogeneity analyses, each of which is driven by a mechanism that is likely to affect diversion but not genuine sales.

6 Tests for Mechanism of BA Devices' Impact

The impact of BA devices on recorded sales due to changes in genuine sales is likely to be different across households depending on their ability to adapt to technology-induced process changes. Similarly, the impact of BA devices on diversion depends on how it affects FPS dealers' payoffs from, and the cost of diverting grains for private benefit. In this section, we estimate the differential impact of BA devices on recorded sales based on FPSs' membership in different categories, where the ability of beneficiaries to adapt to process changes as well as payoffs from and risks of diversion are different across the categories. As suggested by Goldfarb and Tucker (2014), estimating these differential impacts is useful in strengthening causal identification claims. In the following subsections, we estimate differential impact based on the proportion of vulnerable households served by the FPS, FPSs' distance to the nearest market, and prices in the nearest markets to the FPS.

6.1 Proportion of Vulnerable Households Served by the FPS

Prior studies and anecdotalevidence suggest that vulnerable households are more prone to being denied their entitlement by FPS dealers (Sabharwal, 2011; Nagavarapu and Sekhri, 2016; Pradhan and Rao, 2018). This is because vulnerable households, based on their economic and/or social status, have a lower bargaining power reducing their ability to hold the FPS dealer to account. Furthermore, these households find it difficult to adapt to technologically induced process changes owing to their poor educational status and lack of awareness of technology, compared to less vulnerable households.

We expect a negative impact of BA devices on genuine sales to be higher for FPSs that serve a larger proportion of vulnerable households than for those that serve a smaller proportion of such households. On the other hand, the impact of BA devices on diversion would be higher for FPSs that serve a smaller proportion of vulnerable households compared to FPSs that serve a larger proportion of these households, as FPS dealers have greater bargaining power in relation to vulnerable households and can deny entitlements for private gains using the workaround mechanism.

We consider two dimensions of vulnerability—economic and social (based on caste and tribe)—and define the following variables for each FPS.

- We define a binary variable EV_i to indicate if the proportion of economically vulnerable households is high in the population served by FPS i. We set $EV_i = 1$ if the proportion of AAY households served by the FPS is greater than a threshold proportion (90$^{\text{th}}$ percentile of the proportion of AAY households across all FPS–months) and 0 otherwise.
- We define a binary variable SV_i to indicate if the proportion of socially vulnerable households is high in the population served by FPS i. We calculate the proportion of Scheduled Caste (SC) and Scheduled Tribe population in a village (Chatterjee and Sheoran, 2007) using the information at the village level from the Village and Town Amenities Dataset of the district census and set SV_i to 1

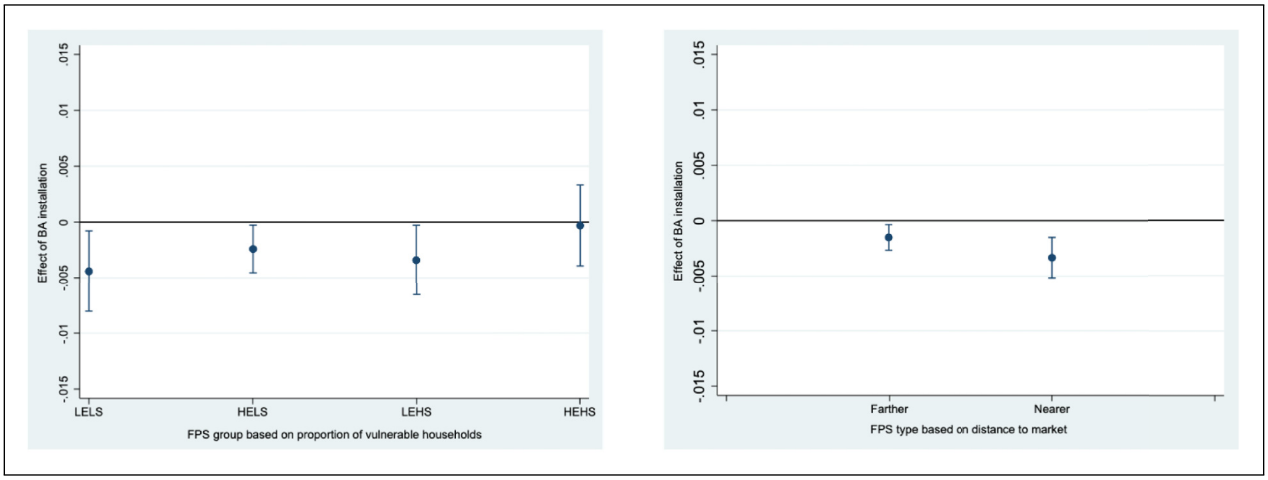

Figure 1. Differential impact of installing BA devices by FPS group based on the proportion of vulnerable households and FPS type based on the distance to market. *Notes*: BA = biometric authentication; FPS = fair price shop. (1) The figure on the left is based on the coefficient estimates of the model specification (5). (2) The *X*-axis indicates different types of FPS based on the proportion of vulnerable households. LELS indicates low proportion of both economic and socially vulnerable households, HELS indicates high proportion of economic and low proportion of socially vulnerable households, LEHS indicates low proportion of economic and high proportion of socially vulnerable households and HEHS indicates high proportion of both economic and socially vulnerable households. The *Y*-axis shows the impact of BA installation along with the 90% confidence intervals. (3) The figure on the right is based on the coefficient estimates of the model specification (5). (4) The *X*-axis indicates different types of FPS based on the distance to market. Nearer indicates nearer to the market and Farther indicates farther to the market. The *Y*-axis shows the impact of BA installation along with the 90% confidence intervals.

if the FPS is located in a village where the proportion of socially vulnerable population is greater than a threshold (90th percentile over all villages in which FPSs are located in our dataset) and 0 otherwise.

We estimate the model below to test our hypothesis on the differential impact of BA devices on diversion by FPSs, based on the proportion of vulnerable households in the population served by the FPS:

$$SGR_{i,t} = \alpha_i + \beta_t + \delta_{ll} \times BA_{i,t} + \delta_{hev} \times EV_i \times BA_{i,t} + \delta_{hsv} \times SV_i \times BA_{i,t} + \delta_{hh} \times EV_i \times SV_i \times BA_{i,t} + \varepsilon_{i,t}. \quad (4)$$

The left panel of Figure 1 shows the impact of BA installation on different FPS groups based on the proportion of vulnerable households. We find that the impact on FPSs with a low proportion of vulnerable households on either or both dimensions, economic and social, are all statistically significant. However, the impact on FPSs with a high proportion of economically and socially vulnerable households is insignificant. Column (3) of Table A3 in the Appendix in the E-Companion shows the results from estimating the above equation. These results suggest that regions with a high proportion of economically and socially vulnerable populations are unable to effectively monitor FPS dealers' behavior, allowing them to use the workaround mechanism (or other mechanisms discussed in Section 3.2) to continue diverting grains.

6.2 FPS Distance to Market

The difference between the market prices (≈25 to 30 INR per kg) and the subsidized PDS price (≈2 to 3 INR per kg) is substantial. For a beneficiary household, this differential in prices is significant and larger than any transaction costs that the use of BA devices is likely to impose. Furthermore, the difference in prices is substantial enough that the relative ease of accessing the market for households in villages closer to the market is unlikely to result in the differential impact of BA devices on genuine sales.

On the other hand, the transaction costs for an FPS dealer are significantly lower in comparison to the benefit from diversion. FPS dealers incur direct, out-of-pocket costs of transportation as well as indirect, opportunity costs of time due to lost business costs to sell the diverted grains in the open markets. FPSs closer to the market are likely to have a lower marginal cost of diversion and therefore divert a larger quantity on average compared to FPSs that are farther away.

BA devices increase the expected cost of diversion and therefore shift the marginal cost (of diversion) curve for all FPSs, leading FPSs to reduce the quantity diverted, everything else remaining the same. However, the impact of the change in marginal cost due to BA devices could lead to a differential impact on diversion for FPSs. The increase in marginal cost due to BA devices is likely to have a lower impact on FPSs that already had a high marginal cost of diversion, as the quantity diverted by such FPSs would be comparatively lower to begin with. We therefore hypothesize that the impact of BA devices on diversion is higher for FPSs that are closer to the open market compared to FPSs that are farther away.

To understand how access to markets affects the impact of BA devices, we focus on the subsample of rural FPSs in our dataset and classify them based on their access to the market. We consider only rural FPSs and distance to the nearest urban market as a proxy for access to the market for the following reasons. First, we do not have precise location information for either FPSs or market locations and therefore have to use proxy measures such as the distance between the village/town that the FPS is located in and the nearest village/town with a market. Second, rural regions have relatively fewer options for the FPS dealers to dispose of the diverted grains while urban markets provide more options and possibly somewhat higher prices. FPS dealers typically sell the diverted grains to retailers or wholesalers in urban areas that are closest to the FPS location. We therefore use the road distance between the village of the rural FPS and the nearest urban center as reported in the Village and Town Amenities Dataset, as a proxy measure for market access. By definition, this proxy measure would be zero for all urban FPSs and market access will not offer variation in the impact of the intervention across urban FPSs, both statistically as well as substantively. Using the above measure of market access for rural FPSs, we have a final dataset of 93,661 FPS–month observations.

We estimate the following triple differences (difference-in-difference-in-differences) model to measure the differential impact by distance to market on rural FPSs:

$$SGR_{i,t} = \alpha_i + \beta_t + \delta_n \times BA_{i,t} + \delta_f \times FAR_i \times BA_{i,t} + \varepsilon_{i,t}, \quad (5)$$

where FAR_i is an indicator variable that takes the value 1 if the distance from a rural FPS i to its nearest urban market is greater than the median distance overall rural FPSs in our data ($\approx$20 km) and 0 otherwise (Iacobucci et al., 2015a, 2015b). The right panel of Figure 1 illustrates the total impact of BA devices on diversion for FPSs that are closer and farther away from the nearest markets, while Column (2) in Table A4 in the Appendix in the E-Companion shows the coefficient values of interest for the above model.[5]

For FPSs that are closer to an open market, installing BA devices reduces diversion by 0.33% of the gross requirement ($\delta_n = -0.0033$ of GR) and the incremental impact on FPSs that are farther away is significant at $p < .1$ and in the opposite direction ($\delta_f = 0.0018$ of GR). The net impact on rural FPSs that are farther away from markets is less than half the estimated impact on FPSs that are nearer, with $\delta_n + \delta_f = -0.0015$ (impact of BA devices is 0.15% of GR) and statistically significant with $p < .05$.

6.3 Market Price for Diverted Grains

FPS dealers dispose of the diverted grains in the nearest open market at the prevailing price in the market. We therefore expect the benefit from diversion to be increasing in the open market price for the grains. FPS dealers (i) do not have significant storage space and thus cannot hold the diverted grains for long, (ii) are price takers in the open market, as the quantity of diverted grains is a very small fraction of the total volume of food grains sold in the open market, and (iii) the commission from distributing to beneficiaries does not vary with market prices. Thus, FPS dealers have a stronger motive to divert using the workaround mechanism (and other means) during periods of high market prices to maximize their payoff. As a result, we hypothesize that BA devices are less effective in curbing diversion at FPSs during periods of high market prices.[6]

We usethe retail price of agricultural commodities collected from various markets periodically by state and central government agencies.[7] The datasets are at the level of a market (village or town), and not at a level of granularity that would be ideal for our analysis. We match the village/town name in the retail price dataset to the name of the village/town in which an FPS is located and use the retail price for the village/town as the open market price at which all FPSs located in the village/town can dispose of the diverted grains. The retail price data is available only for 112 villages across two districts, which correspond to 25,885 FPS–month observations in our dataset.

We note that the variation in market prices is much lower than the difference between the retail price and the commission paid to FPS dealers in the PDS, as seen in Figure A3 in the Appendix in the E-Companion, which show box plots of market prices across markets for different months. The average retail price of rice for the markets in our data during our analysis period is 34.40 INR per kg and the standard deviation is 2.97 INR yielding a coefficient of variation (COV) of 0.08. The FPS dealers were paid a commission at the rate of $\approx$1 INR per kg of grains distributed to eligible beneficiaries (Correspondent, 2018).

Given the limited variation in the market prices in our data, we use a median split model to measure the differential impact of BA devices based on market price using the following specification:

$$SGR_{i,t} = \alpha_i + \beta_t + \delta \times BA_{i,t} + \gamma_m \times HI\_RET_{i,t} + \delta_m \times HI\_RET_{i,t} \times BA_{i,t} + \varepsilon_{i,t}, \quad (6)$$

where $HI\_RET_{i,t}$ takes value 1 if the retail price for the FPS i in month t is greater than the median retail price overall FPS–month observations in our data and 0 otherwise.

As a robustness check, we also estimate the differential impact as a function of market price using:

$$SGR_{i,t} = \alpha_i + \beta_t + \delta \times BA_{i,t} + \gamma_m \times RET_{i,t} + \delta_m \times RET_{i,t} \times BA_{i,t} + \varepsilon_{i,t}, \quad (7)$$

where $\mathrm{RET}_{i,t}$ denotes the retail market price for FPS i in month t, and is equal to the price prevailing in the village or town in which FPS i is located in month t.

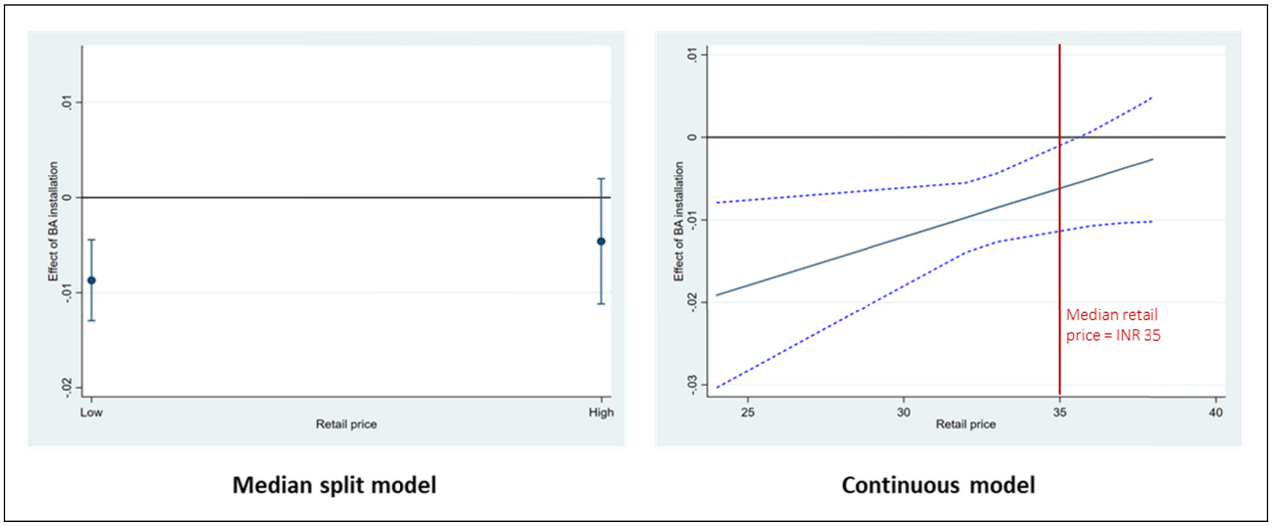

Figure 2. Impact of biometric authentication (BA) devices at different levels of retail price. *Notes*: The figure on the left shows the results of estimating equation (6). The figure on the right shows the results of estimating equation (7) at different values of the retail price. The *X*-axis indicates the market price and the *Y*-axis shows the impact along with 90% confidence intervals.

We reestimate our original specification equation (3) on this curtailed dataset to compare the results with the estimates from equations (6) and (7). The results of estimating equations (6) and (7) are shown in Figure 2 while the values of the relevant coefficients are given in Table A5 in the Appendix in the E-Companion.

For the median split model (6), we find that the net impact of BA devices is statistically significant for FPS–months when the retail price is less than the median value ($\delta = -0.0087$) but not so when they are greater than the median value ($\delta + \delta_m = -0.0047$). In the specification where the retail price is modeled as a continuous value, (7), the coefficient of the interaction term, δ_m, itself is not statistically significant. However, $\delta + \delta_m \times \mathrm{RET}_{i,t}$ is increasing in the retail price and statistically significant for $\mathrm{RET}_{i,t} \leq 36$. The coefficient $\delta + \delta_m \times \mathrm{RET}_{i,t}$ becomes statistically insignificant for $\mathrm{RET}_{i,t} >$ INR 36, which is close to the median price overall FPS–month combinations in our dataset. These results suggest that BA installation is able to reduce diversion when retail prices are low but not otherwise.

The differential impact of BA devices on FPSs based on the proportion of vulnerable households served, their distance to market, and open market prices suggest that the underlying mechanism through which they affect recorded sales is through their impact on diversion rather than genuine sales to beneficiaries. Taken together, the results of our differential impact analyses offer valuable managerial insights. We find that regions with a high proportion of economically and socially vulnerable populations are unable to effectively monitor FPS dealers' behavior, allowing them to exploit workaround mechanisms for diverting grains. Additionally, we find that the impact of BA installation is influenced by market access and that BA installation is able to reduce diversion when retail prices are low but not otherwise. Overall, we find that the impact of BA devices is lower in regions that have a greater proportion of extremely vulnerable households, rural regions that are farther away from closest urban areas, and in regions and periods when market prices for food grains are high. A direct prescription for policymakers is to prioritize such areas by conducting targeted audits at FPSs in rural areas and regions and periods where market prices are high.

7 Value of Better Planning Using Timely Inventory Information

As discussed in Section 3.1, lagged sales and inventory information captured manually at the FPSs was used to create a centralized plan of inventory replenishments to FPSs. This can lead to suboptimal demand–supply matching due to imprecise information, even in a centrally planned supply chain such as the PDS. The BA devices in our context were equipped with the additional capability of recording real-time information on sales and inventory at the FPSs. This information captured by the BA devices can be utilized by central planners to improve replenishment planning and create additional value. We term this additional as value of better planning.

In our study setting, the Government of Karnataka continued to use lagged inventory information to determine replenishment quantities even after the implementation of BA devices. As a result, we cannot quantify the value of better planning empirically. To overcome this, we conduct an extensive simulation study to model the replenishment planning process if information captured by BA devices is used and quantify the value of better planning. We model various scenarios, with different values of key input parameters that are representative of the characteristics of the PDS in various Indian states.

Next, we describe our simulation study design (Section 7.1), choice of input parameters (Section 7.2) followed by the discussion of our results and associated insights (Section 7.3).

7.1 Simulation Design

To quantify the incremental value of better planning, we compare the performance of the PDS supply chain under three scenarios: (i) Baseline, which simulates the planning and monitoring processes in the absence of BA devices, (ii) Monitoring, which simulates improved monitoring (reduced diversion) due to BA devices but unchanged planning process, and (iii) Planning, which simulates improved planning (calculation of replenishments using timely inventory information) in addition to improved monitoring.

In the Baseline scenario, for each FPS $i \in \{1, 2, \dots, I\}$ and month $t \in \{1, 2, \dots, T\}$, we simulate genuine demand for food grains from beneficiaries and the quantity of grains that the FPS dealer intends to divert as proportions of the gross requirement. We denote these by $D_{i,t} = p_{i,t} \times \mathrm{GR}_{i,t}$ and $\hat{L}^{(b)}_{i,t} = l^{(b)}_{i,t} \times \mathrm{GR}_{i,t}$, respectively. We assume that $p_{i,t}$ and $l^{(b)}_{i,t}$ follow a uniform distribution with support on subsets of (0, 1). The replenishment quantity for FPS i and month t is calculated as $Q^{(b)}_{i,t} = \mathrm{GR}_{i,t} - \mathrm{CB}^{(b)}_{i,t-2}$ denoting the use of lagged inventory information under the Baseline scenario. This brings the inventory of food grains available at the beginning of month t to $\mathrm{OB}^{(b)}_{i,t} = \mathrm{CB}^{(b)}_{i,t-1} + Q^{(b)}_{i,t}$.[8]

We do not have granular operational data on the temporal coevolution of the demand fulfillment (DF) and diversion of grains (DG) at the FPS. We consider two extreme cases to obtain upper and lower bounds on the value of better planning. Under the DF $\rightarrow$ DG assumption, sale to genuine beneficiaries occurs first, followed by the diversion of grains. The sales to genuine beneficiaries is given by $\hat{S}^{(b)}_{i,t} = \min\{D_{i,t}, \mathrm{OB}^{(b)}_{i,t}\}$, and quantity of grains diverted is equal to $L^{(b)}_{i,t} = \min\{\hat{L}^{(b)}_{i,t}, \mathrm{OB}^{(b)}_{i,t} - \hat{S}^{(b)}_{i,t}\}$. In contrast, under the DG $\rightarrow$ DF assumption, the FPS dealer diverts grains first, before disbursing grains to genuine beneficiaries who come to collect their entitlements. The quantity of grains diverted is given by $L^{(b)}_{i,t} = \min\{\hat{L}^{(b)}_{i,t}, \mathrm{OB}^{(b)}_{i,t}\}$, while sales to genuine beneficiaries is given by $\hat{S}^{(b)}_{i,t} = \min\{D_{i,t}, \mathrm{OB}^{(b)}_{i,t} - L^{(b)}_{i,t}\}$ in this case. For both DF $\rightarrow$ DG and DG $\rightarrow$ DF, the recorded sales is a sum of genuine sales and diversion and is given by $S^{(b)}_{i,t} = \hat{S}^{(b)}_{i,t} + L^{(b)}_{i,t}$, while the ending inventory is given by $\mathrm{CB}^{(b)}_{i,t} = \mathrm{OB}^{(b)}_{i,t} - S^{(b)}_{i,t}$.

Under the Monitoring scenario, all calculations are similar to those described above except for the quantity of grains that the FPS dealer intends to divert. It is given by $\hat{L}^{(m)}_{i,t} = l^{(m)}_{i,t} \times \mathrm{GR}_{i,t}$, where $l^{(m)}_{i,t}$ follows a uniform distribution whose mean is lower than that of $l^{(b)}_{i,t}$ to reflect the value of better monitoring (lower diversion of grains) due to BA devices. Similarly, all calculations in the Planning scenario are identical to that in the Monitoring scenario except that the replenishment quantity is calculated using updated inventory information and is given by $Q^{(p)}_{i,t} = \mathrm{GR}_{i,t} - \mathrm{CB}^{(p)}_{i,t-1}$.

We use the results from the above simulation to calculate two operational outcomes: (a) average reduction in the inventory levels,

$$\frac{\sum_{i,t} \left(CB^{(m)}_{i,t} - CB^{(p)}_{i,t} \right)}{I \times T}$$

and (b) average increase in genuine sales to beneficiaries,

$$\frac{\sum_{i,t} \left(\hat{S}^{(p)}_{i,t} - \hat{S}^{(m)}_{i,t} \right)}{I \times T}$$

because of better planning. We estimate the incremental monetary value from better planning by calculating the reduction in inventory holding cost for the PDS and financial cost of open market purchases for households and characterize their sum as the incremental value of better planning. We also calculate the value of better monitoring as an average decrease in diversion,

$$\frac{\sum_{i,t} \left(L^{(b)}_{i,t} - L^{(m)}_{i,t} \right)}{I \times T},$$

and subsequently the ratio of the incremental value of better planning (reduced inventory and reduced lost sales) and the value of better monitoring (reduced diversion) estimated from the simulation model to compare their magnitudes.

7.2 Simulation Parameters

For each of the 6,934 FPSs ($i = 1, \dots, 6,934$) in our dataset described in Section 4.1, we initialize the simulation model with ending inventory values for the first two months $t = 1, 2$. We then use these in conjunction with the gross requirements for 28 months to simulate the inventory, sales, and diversion dynamics for subsequent months $t = 3, \dots, 28$ as described in Section 7.1.

We choose different supports for demand ($p_{i,t}$) and diversion in the Baseline scenario ($l^{(b)}_{i,t}$) such that all feasible combinations of their means approximately match the high, medium, and low levels of the reported values of PDS uptake and diversion across Indian states (Khera, 2011; NCAER, 2015; Dreze and Khera, 2015; Gulati and Saini, 2015).

In particular, we consider all feasible combinations of $\mathbf{E}(p_{i,t}) \in \{0.45, 0.75, 0.90\}$ and $\mathbf{E}(l^{b}_{i,t}) \in \{0.05, 0.20, 0.40\}$ such that $p_{i,t} + l^{b}_{i,t} \leq 1$. For each combination, we choose the support for $l^{(m)}_{i,t}$ such that the difference in the average quantity actually diverted between the Baseline and Monitoring scenarios,

$$\frac{\sum_{i,t} \left(L^{(b)}_{i,t} - L^{(m)}_{i,t} \right)}{I \times T},$$

is approximately equal to the impact of BA devices.

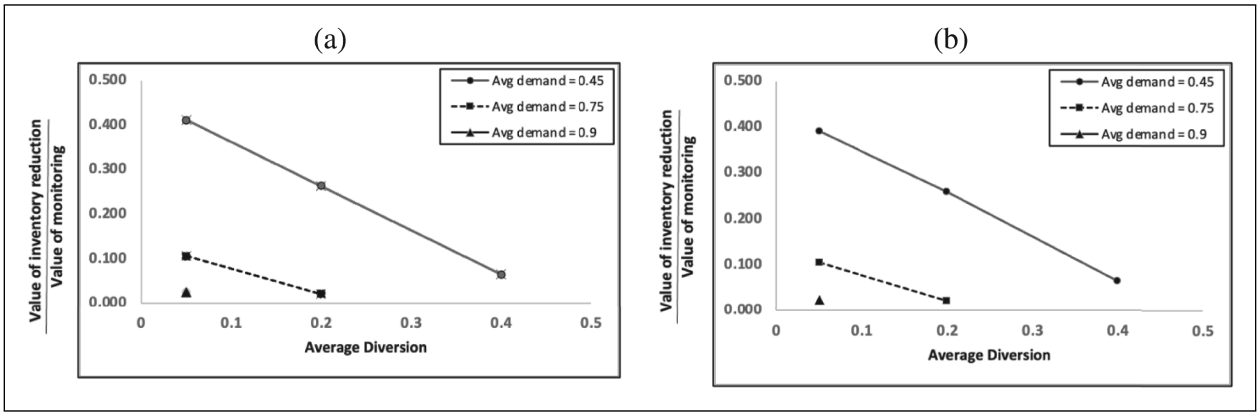

Figure 3. *Value of better planning* in terms of lower inventory cost relative to value of better monitoring for different levels of demand and diversion: (a) (DF → DG); (b) (DG → DF.) *Notes*: (1) The *X*-axis corresponds to the average diversion level under the Baseline scenario while the different lines correspond to different average demand from genuine beneficiaries, with diversion and demand expressed as a fraction of the gross requirement. (2) The *Y*-axis indicates the value from lower inventory holding cost as a fraction of the value from better monitoring for different (feasible) combinations of average demand and diversion levels. DF → DG denotes demand fulfillment precedes diversion of grains while DG → DF denotes diversion of grains precedes demand fulfillment.

We assume an inventory holding cost of 8% per year and an open market price of INR 28 per kg of rice to compute the monetary value of operational improvements.[9] For each combination of input parameters, scenario, and sequence of events, we conduct 1,000 simulation runs and report the average value of the performance metrics.

7.2.1 Calibration of the Simulation Model. We calibrate the model to ensure that the temporal variation in CB in the simulation aligns closely with the temporal variation in CB observed in the dataset. We achieve this calibration by varying the supports (parameters) of the uniform distribution of both the demand and diversion parameters ($p_{i,t}$ and $l_{i,t}$), while keeping the average values as described earlier. The adjustment of these distribution parameters impacts the variability of the random variable. This variation in the demand and diversion parameters subsequently reflected in the variation of CB. We carefully choose the distribution parameters such that the CV of CB and serial correlation of CB values in the simulation, for the monitoring scenario, for all combinations of demand and diversion and for both the assumptions (DF → DG and DG → DF) closely match the respective values observed in the empirical dataset (CV of CB = 1.05 and serial correlation of CB = 0.174 in the dataset).[10] Subsequently, we employ these calibrated parameters to rerun the entire simulation study. The actual values of CV and serial correlation in the simulation after calibration are shown in Table A13 in the Appendix in the E-Companion.

7.3 Simulation Results

We analyze the value from reduced inventory holding costs and that from reduced lost sales separately, to better understand the mechanisms.[11]

Figure 3 shows that, for a given level of demand (diversion), the benefit from reduced inventory is decreasing in Baseline scenario diversion (demand). All else being equal, a greater sum of demand and diversion results in ending inventory being closer to zero in consecutive time periods and displaying lower variability across periods. In such a situation, that is, if $|\mathrm{CB}_{i,t-2} - \mathrm{CB}_{i,t-1}|$ is small, using lagged ($\mathrm{CB}_{i,t-2}$) or updated ($\mathrm{CB}_{i,t-1}$) inventory information (corresponding to Monitoring and Planning scenarios) results in similar replenishment quantities. Since this mechanism is driven almost exclusively by the sum of demand and diversion in Baseline scenario, the magnitude of this effect is not different between DF → DG and DG → DF assumptions. To demonstrate the underlying mechanism more clearly, we plot the value from reduced inventory for each combination of demand and diversion against the average value of ($\max_t \mathrm{CB}_{i,t} - \min_t \mathrm{CB}_{i,t}$) in the Monitoring scenario (Figure A8 in the Appendix in the E-Companion). As expected, we observe a monotone behavior between the two, which is almost identical for both DF → DG and DG → DF assumptions.

Figure 4 shows that the benefit from reduced lost sales is nonmonotonic in the extent of diversion (demand) for low levels of demand (diversion). To understand this observation, consider the following two extreme scenarios. On the one hand, when the sum of demand and diversion is high, as explained above and seen in Figure A8 in the Appendix in the E-Companion, variation in ending inventory across periods is low. Hence, the value obtained from a reduction in lost sales is low too. On the other hand, when demand and diversion are both low, although there is greater variation in ending inventory across periods, the probability of occurrence of lost sales itself is low due to low levels of demand. Hence the value of the reduction in lost sales is again low. For intermediate levels

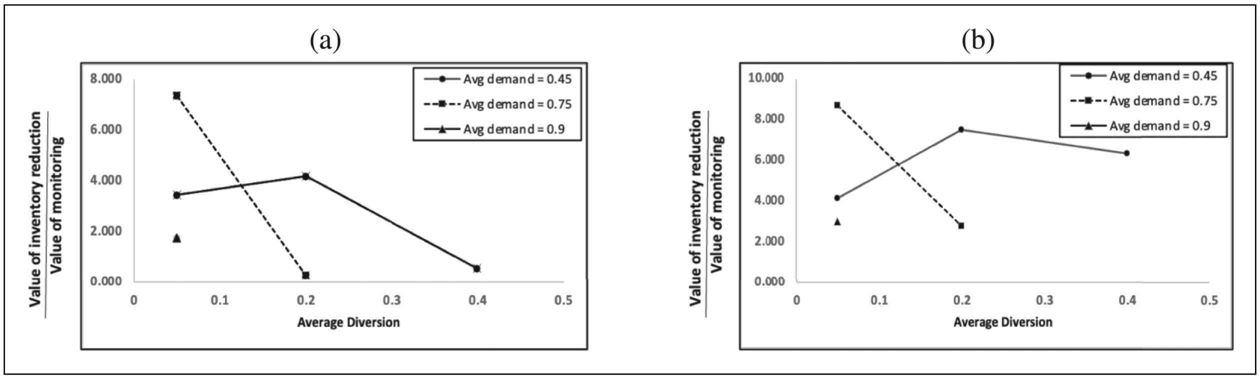

Figure 4. *Value of better planning* in terms of lower lost sales cost relative to value of better monitoring for different levels of demand and diversion: (a) (DF → DG) and (b) (DG → DF). *Notes*: (1) The *X*-axis corresponds to the average diversion level under the Baseline scenario while the different lines correspond to different average demand from genuine beneficiaries, with diversion and demand expressed as a fraction of the gross requirement. (2) The *Y*-axis indicates the value to beneficiaries from a reduction in lost sales as a fraction of the value from better monitoring for different (feasible) combinations of average demand and diversion levels. DF → DG denotes demand fulfillment precedes diversion of grains while DG → DF denotes diversion of grains precedes demand fulfillment.

of demand and diversion, both variability in ending inventory across periods as well as the probability of occurrence of lost sales are sizeable. Hence, the value obtained from reduced lost sales is high under these scenarios. Finally, comparing Figures 4a and 4b shows that the benefit from reduced lost sales due to better planning is higher for DG → DF as a shortage in availability has a greater impact on sales to genuine beneficiaries under this sequence.

For the sake of clarity and simplicity in communicating the overall benefits of planning, we aggregate the two components (benefit from reduced inventory and benefit from reduced lost sales). We believe that this combined approach offers a more parsimonious representation of the total value derived from planning, encompassing benefits related to inventory carrying costs and lost sales. Table 3 shows the ratio of value of better planning and value of better monitoring for various combinations of demand and diversion values under DF → DG and DG → DF cases. This ratio ranges from around 0.31 to around 8.82 depending on the magnitude of the genuine demand from beneficiaries and the extent of diversion in the Baseline scenario and is nonmonotone with respect to demand and diversion. The value of better planning is almost equal to thrice that of the value of better monitoring on average for the DF → DG case whereas it is six times as much for DG → DF.

We also find, somewhat paradoxically, that the actual quantity diverted increases under the Planning scenario (compared to the Monitoring scenario) under both DF → DG and DG → DF cases. Figure A9 in the Appendix in the E-Companion shows the percentage increase in actual diversion between planning and monitoring scenarios for different levels of demand and diversion. The average increase in diversion between Monitoring and Planning scenarios is 12.7% under DF → DG and 0.33% under DG → DF (as % of quantity diverted under the Monitoring scenario). Better planning results in better availability of food grains as seen from the increase in sales to genuine beneficiaries (Figure 4). This also means on average there is a higher quantity of grains available for the FPS dealer to divert when updated information is used to plan replenishments. Under the DF → DG assumption, the FPS dealer diverts grains after fulfilling demand. Better availability of grains results in higher quantity available after demand fulfillment on average, resulting in higher quantity being diverted under the DF → DG case. Under the DG → DF case, diversion occurs before demand fulfillment. As the total quantity diverted as a fraction of GR is small (≈10%), the improved availability due to better planning has a lesser impact on the quantity diverted in this case.

7.4 Sensitivity Analysis

7.4.1 Monitoring Effectiveness of BA Devices. We replicate the simulation for two levels of impact of BA devices. (i) 0.00195 (ii) 0.0078, corresponding to half and double our empirical estimate (0.0039). Figure A10 in the Appendix in the E-Companion shows the average value from planning across different (feasible) combinations of average demand and diversion levels for different levels of impact of BA devices. As expected, as the impact of BA devices increases, the ratio of value from better planning to value from monitoring decreases, and conversely, when the value from monitoring decreases, the ratio rises for both DF → DG and DG → DF scenarios.

7.4.2 Holding Costs. We also conduct sensitivity analysis by allowing the holding cost assumption to vary within a range of 6% to 10%. This modification is reflected in Figure A11 in the Appendix in the E-Companion, where bands denoting

Table 3. *Value of better planning* relative to *value of better monitoring* for different levels of demand and diversion.

Average diversion	Average demand	*Value of better planning as a fraction of Value of better monitoring*	
		DF → DG	*DG → DF*
0.05	0.45	3.88	4.54
0.05	0.75	7.46	8.82
0.05	0.90	1.79	3.04
0.20	0.45	4.47	7.79
0.20	0.75	0.31	2.83
0.20	0.90	NA	NA
0.40	0.45	0.64	6.42
0.40	0.75	NA	NA
0.40	0.90	NA	NA
	Average	**3.09**	**5.57**

Notes: (1) Average demand and average diversion are expressed as a proportion of gross requirement. (2) Average demand refers to the demand from genuine beneficiaries while average diversion refers to the level of diversion in the Baseline scenario. (3) NA denotes that the particular combination of demand and leakage values is not possible as it exceeds 100% of gross requirement. (4) DF → DG denotes demand fulfillment precedes diversion of grains while DG → DF denotes diversion of grains precedes demand fulfillment.

6% and 10% holding costs have been added around the initial 8% holding cost assumption. We find that the average ratio of the value of lower inventory to the value of monitoring varies between 0.11 and 0.19 for 6% and 10% holding costs with a value of 0.15 for 8% holding costs under the assumption of DF → DG. Similarly, the average ratio of the value of lower inventory to the value of monitoring varies between 0.11 and 0.18 for 6% and 10% holding costs with a value of 0.14 for 8% holding costs under the assumption of DG → DF. Also, note that the value from decreased lost sales is much higher (average value of 2.94) compared to the value from reduced inventory (average value of 0.15), and hence a slight variation in the assumption related to holding costs does not drastically change the overall benefits from planning.

8 Discussion

Our results indicate that policymakers should expect a limited overall impact of technological interventions such as the use of BA devices in low- and middle-income countries due to specific contextual factors. In particular, weak technological infrastructure necessitates the use of workaround mechanisms along with technology such as BA devices. However, asymmetric power dynamics between intermediaries (government- appointed agents who manage) and beneficiaries can lead to misuse of the workaround mechanism, leading to the limited impact of technological interventions on improving performance. Factors that enhance (or mitigate) the impact of BA devices in our setting are likely to be present to varying degrees in other settings too and our findings can be used by managers and policymakers to drive decisions that can improve the effectiveness of BA devices in their context.

Our finding that BA devices have a low impact on diversion, in the presence of workaround mechanisms is particularly crucial in developing economies where seamless implementation of technological interventions is not yet prevalent. The existence of workaround mechanisms becomes important to safeguard genuine beneficiaries. Hence, managers in such economies seeking to implement similar interventions can leverage our findings to make informed decisions. For instance, they could ask the FPS dealers to document transaction-specific reasons for utilizing workaround mechanisms. While this entails additional time and effort, it adds a layer of complexity to potential misuse by FPS owners compared to allowing unexplained workaround usage. Policymakers can also weigh the welfare loss of genuine beneficiaries against the utilization of workaround mechanisms. Policymakers should design workaround mechanisms to strike a balance between misuse by intermediaries for their private benefit and interrupted provision of entitlements to beneficiaries. For instance, the effectiveness of BA devices in curbing diversion may be improved if they are accompanied by a targeted auditing of FPSs that record a higher volume of transactions using the workaround mechanism.

We also find that the use of the workaround mechanism (and hence the impact of BA devices) varies depending on the characteristics of the households, FPSs, and prevailing market conditions. We find that the impact of BA devices is lower in regions that have a greater proportion of extremely vulnerable households, rural regions that are farther away from the closest urban areas, and in regions and periods when market prices for food grains are high. When taken together, the results from Section 6, which demonstrate differential impact across various dimensions, validate the use of recorded sales as a credible proxy for diversion.

These results highlight although new digital technologies intend to benefit the vulnerable sections of the society, they are less likely to accrue those benefits because of their lack of familiarity with technology. Policymakers must ensure that the implementation of technology in public welfare programs is accompanied by strong publicity and awareness campaigns targeted toward the most vulnerable with the aim of overcoming barriers to accessing digital technologies. Policymakers can improve the effectiveness of interventions such as BA devices by targeting regions and periods where market prices are high, which also are regions and periods when the reliance on welfare programs is highest, for additional auditing and making better use of the limited human resources.

Developing countries such as Indonesia, Mexico, Egypt, and Sri Lanka run food subsidy programs with in-kind transfer of grains to beneficiaries and suffer from similar problems such as leakages, inclusion–exclusion errors and errors related to beneficiary identification (Abdalla and Al-Shawarby, 2017; Alderman et al., 2018, 2017; Tilakaratna and Sooriyamudali, 2017; Timmer et al., 2017; APEC, 2017) and are turning toward technology to combat corruption (Wickberg, 2013; Silveira, 2016; Santiso, 2019; Mukherjee, 2015). For instance, Mexico recently launched the first biometric card trial in the state of Sonora for reducing fraud in the disbursement of state benefits (Stephen, 2019; Cuttler , 2019). Our findings can be used by managers and policymakers involved in these programs to drive decisions that can improve the effectiveness of technology-based interventions.

Given that the impact of BA devices on monitoring itself is limited and heterogeneous, it is of utmost importance that the policymakers take advantage of the real-time information collected from these devices to improve operations and supply chain planning, especially in programs that involve the physical delivery of goods. In our context, ignoring the additional benefit of better planning and focusing only on the value of better monitoring may severely underestimate the value of such BA devices in other welfare programs that experience diversion in manual paper-based systems (e.g., Barnwal, 2014; Muralidharan et al., 2016).

Our extensive simulation exercise identifies how the incremental benefit from using information captured by BA devices for planning varies based on the existing levels of diversion and genuine demand, which can be used by policymakers to prioritize their efforts.

State authorities can use the simulation results to evaluate the anticipated benefits from planning, given estimates for actual demand and diversion in their states against the potential costs associated with recording, storing, and processing data collected by BA devices to formulate replenishment policies. Moreover, policymakers can use findings from our sensitivity analyses to estimate the incremental value from planning under a more stringent workaround mechanism in anticipation of a greater monitoring value from BA devices.

9 Conclusion

We employ an operational lens to evaluate technology-based interventions to improve the performance of public welfare delivery supply chains. While we find that the installation of BA devices had a statistically significant impact on the diversion of grains, we also recognize that, in our context, the digital technology of BA has not proven to be as effective as expected, primarily due to the presence of the workaround mechanism. As a result, the full potential benefits of such technology have not been fully realized and the limited impact that we find is a result of the contextual factors of our study setting.

Our analysis has limitations due to a lack of data availability. We are unable to empirically link the usage of the workaround mechanism (paper-based method) and the extent of diversion, as data on transactions conducted using the workaround mechanism is not available. We also do not have data on the unavailability of BA devices due to a lack of data on the frequency of breakdown of BA devices installed in the FPSs. Moreover, we also do not have data on the number of auditors and their field activities, rendering empirical analysis of manual monitoring mechanisms infeasible.

The context studied in the current paper provides avenues for future research in terms of developing both empirical and analytical models. A basic analytical model could capture the tradeoffs influencing the diversion decisions of FPS dealers and quantify the impact of BA devices. This formulation could be used to answer policy questions. First, it could elucidate additional potential mechanisms underlying our empirical analysis results. Second, exploring the formulation and analysis of a comprehensive model that captures the strategic behavior of an FPS dealer based on private information, coupled with designing interventions to improve the effectiveness of BA devices, presents an interesting avenue for future analytical research. Finally, the information captured by BA devices is a distorted signal of actual sales to beneficiaries. This is because the recorded sales are a combination of actual sales and diversion. The problem of estimating the true demand signal from a distorted signal such as recorded sales and how it can be used for planning is another interesting avenue for future empirical research.

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ORCID iD

Maya Ganesh https://orcid.org/0000-0002-4231-6893

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Notes

1. We describe the details of the data cleaning procedure in Appendix B in the E-Companion and the summary statistics of the data including outliers is given in Table A1 in the Appendix in the E-Companion.
2. Details of the verification of parallel trends, a key identifying assumption in DID estimation, for our model are given in Appendix D in the E-Companion. Appendix E in the E-Companion has details of the review of news and media articles to ascertain that no other significant events that would bias the estimates occurred during the study period. Our results are robust to several alternative inclusion/exclusion criteria for the outliers, as shown in Table A2 in the Appendix in the E-Companion and continue to hold if the errors are clustered at the district and month level. We also estimated the above model using total recorded sales for rice and wheat and found a similar effect (0.42% of the gross requirement). However, we found no significant effect on the diversion of wheat. This could be due to smaller quantities of wheat distributed through PDS compared to rice, which may limit the statistical variation required for identification.
3. Economic cost of procurement, storage, transportation, and distribution of grains during the period of analysis was INR 27.02 per kg of rice as per statistics published by the Government of India in the Food Grain Bulletin. See https://dfpd.gov.in/Home/ for details.
4. We also calculate the standard errors and 90% confidence interval for the savings which we compute using the standard error of point estimate of the impact of BA devices. For the 3,075 FPSs that received BA devices in our analysis data set, the reduction in diversion translates to 111.77 MT per month (standard error of 52.77 MT and 90% confidence interval of 22.18–202.45). In monetary terms, the reduction in the procurement requirements translates to INR 36.3 million per year (standard error of INR 17.10 million and 90% confidence interval of INR 7.19 million and INR 65.64 million).
5. Our results are robust to alternate definitions of FAR_i as being greater than the 60^{th} and 75^{th} percentile of distance between the rural FPS and its nearest urban market (25 and 31 km, respectively). The results are shown in rows (3)–(6) of Table A4 in the Appendix in the E-Companion.
6. Household consumption of food grains is relatively inelastic to price changes and there is unlikely to be any variation in a household's use of the open market as a substitute to the PDS. Thus, there is unlikely to be a differential impact of BA devices on genuine sales.
7. The retail price data is published by the Directorate of Economics and Statistics and is available at https://rpms.da.gov.in/
8. OB stands for opening balance and CB stands for closing balance.
9. For the cost of capital, we use the average term deposit bank interest rates of State Bank of India during our study period. See https://www.sbi.co.in/portal/web/interest-rates/old-interest-rates-last-10-years for details. For the open market price, we use the average wholesale market price of rice in Karnataka during our study period, available on the AGMARKNET portal which collects, analyses, and disseminates market information to farmers, traders, and policymakers. See http://agmarknet.gov.in/Default.aspx for details.
10. We use the CV as it provides a normalized measure of variation that accounts for differences in scale and magnitude of different FPSs. This metric, being dimensionless, simplifies comparisons by accounting for the relative variability between FPSs. Furthermore, to ensure an appropriate comparison with the monitoring scenario, we compute the empirical CV exclusively for those observations in the empirical dataset that have BA installation.
11. As expected, the value from monitoring is similar between the empirical estimation and the simulation study; we find that the value (kg per FPS per month) varies between 34.98 and 37.11 in the simulation across different scenarios as compared to 36.3 from the empirical estimation.

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